Executive Summary

Project Overview

This project presents an Exploratory Data Analysis (EDA) of an E-Commerce Sales dataset to evaluate sales performance, customer purchasing behaviour, profitability, regional performance, product trends, and seasonal patterns. The dataset was cleaned by handling missing values, duplicates, data type inconsistencies, logical validation, and outliers before analysis. Python, Pandas, NumPy, Matplotlib, and Seaborn were used to perform the complete analysis.

Key Findings

The analysis identified important trends in sales, profit, customer segments, product categories, regional performance, and time-based behaviour. Sales and profit exhibited a strong positive relationship, while customer and regional analyses highlighted valuable business opportunities. Seasonal analysis revealed periods of higher demand that can support better planning and inventory management.

Business Insights

The results indicate that focusing on profitable product categories, improving customer retention, optimizing promotional strategies, and strengthening lower-performing regions can improve business performance. Seasonal demand patterns should also be considered while planning marketing campaigns and inventory.

Conclusion

This project demonstrates how Exploratory Data Analysis can transform raw transactional data into meaningful business insights. The findings support data-driven decision-making and provide practical recommendations that can help improve profitability, operational efficiency, and overall business growth.

Recommendations

- Increase investment in high-performing product categories.
- Strengthen customer retention through loyalty initiatives.
- Apply targeted discounts instead of blanket promotions.
- Improve sales performance in lower-performing regions.
- Plan inventory using seasonal sales trends.
- Monitor business KPIs regularly using dashboards.